

Ten years of life: Is it a matter of choice?

BACKGROUND: Relative risk estimates suggest that effective implementation of behaviors commonly advocated in preventive medicine should increase life expectancy, although there is little direct evidence. **OBJECTIVE:** To test the hypothesis that choices regarding diet, exercise, and smoking influence life expectancy. **METHODS:** A total of 34 192 California Seventh-Day Adventists (75% of those eligible) were enrolled in a cohort and followed up from 1976 to 1988. A mailed questionnaire provided dietary and other exposure information at study baseline. Mortality for all subjects was ascertained by matching to state death tapes and the National Death Index. **RESULTS:** California Adventists have higher life expectancies at the age of 30 years than other white Californians by 7.28 years (95% confidence interval, 6.59-7.97 years) in men and by 4.42 years (95% confidence interval, 3.96-4.88 years) in women, giving them perhaps the highest life expectancy of any formally described population. Commonly observed combinations of diet, exercise, body mass index, past smoking habits, and hormone replacement therapy (in women) can account for differences of up to 10 years of life expectancy among Adventists. A comparison of life expectancy when these factors take high-risk compared with low-risk values shows independent effects that vary between 1.06 and 2.74 years for different variables. The effect of each variable is assessed with all others at either medium- or high-risk levels. **CONCLUSIONS:** Choices regarding diet, exercise, cigarette smoking, body weight, and hormone replacement therapy, in combination, appear to change life expectancy by many years. The longevity experience of Adventists probably demonstrates the beneficial effects of more optimal behaviors.